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Service member for a vehicle cooling module and a vehicle cooling module with the service member

Abstract

A servicing member for a vehicle cooling module includes a first element with a bumper portion, a first fixing interface located on the first element for fixing to the vehicle module, a second element connected to the first element and a second fixing interface located on the second element for fixing the servicing member to a vehicle structure. The first element is connected with the second element by a connection portion adapted to controllably break upon imparting a predetermined force to the bumper portion. The first fixing interface includes a tongue portion adapted for releasable connection with a slot in the vehicle cooling module, the first fixing interface including a first side wall extending adjacent to the tongue portion between said tongue portion and the bumper portion so that a free space is created between the tongue portion and the first side wall.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is filed under 35 U.S.C. § 371 U.S. National Phase of International Application No. PCT/EP2021/059295 filed Apr. 9, 2021 (published as WO2021213816), which claims priority benefit to European Application No. 20461530.6 filed on Apr. 20, 2020, the disclosures of which are herein incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The invention concerns a service member for replacing a broken mounting assembly of a vehicle cooling module. The invention further relates to a vehicle cooling module with a service member and a combination thereof.

BACKGROUND OF THE INVENTION

(3) Vehicles of all kinds, for example motor vehicles, passenger cars and trucks, have cooling components for a cooling system. Usually, cooling radiators are located at the front of the vehicle, behind the bumper. These are therefore often damaged during a head-on collision, even at low speeds. If the water tank or the core of the cooling element is damaged, the radiator is out of order and the vehicle cannot be driven to the nearest repair shop and must be towed away. In addition, if the water tank or, more generally, the cooling radiator is damaged, the radiator must be replaced with a new radiator, resulting in high repair costs.

(4) It is already known from the state of the art to mount the radiator or more generally the vehicle cooling module on the vehicle with mounting assemblies. In the event of a frontal collision, these mounting assemblies break off and the radiator itself remains undamaged. After an accident, the mounting assembly must be replaced.

(5) In known solutions, the vehicle cooling module is fitted with a separate plastic fixing element, e.g. a dovetail on successive cooling modules. In the event of an accident, this separate plastic fixing element is replaced by an identical part. This solution requires an additional part, i.e. the separate plastic fixing element, to be attached not only to one vehicle cooling module as a repair solution after an accident, but to all successive cooling modules. This means additional working time and thus additional costs.

(6) There is a need for a service member, which can be easily attached to a vehicle cooling module after an accident, thus minimizing the costs for the repair. There is also a need for a vehicle cooling module that is mounted in such a way that in the event of a head-on collision the water tank will not leak and that is easy to repair with a service member.

BRIEF SUMMARY OF THE INVENTION

(7) The object of the invention is, among others, a servicing member for a vehicle cooling module, comprising a first element with a bumper portion, a first fixing interface located on the first element for fixing to the vehicle module, a second element connected to the first element, a second fixing interface located on the second element for fixing the servicing member to a vehicle structure, wherein the first element is connected with the second element by a connection portion adapted to controllably break upon imparting a predetermined force to the bumper portion, wherein the first fixing interface comprises a tongue portion, the tongue portion being adapted for releasable connection with a slot in the vehicle cooling module, the first fixing interface comprising a first side wall extending adjacent to the tongue portion between said tongue portion and the bumper portion so that a free space is created between the tongue portion and the first side wall.

- (8) Preferably, wherein the tongue portion is of plate-like shape and rectangular cross-section.
 - (9) Preferably, the tongue portion comprises a blocking section adapted for frictional fixing in a slot of a vehicle cooling module.
 - (10) Preferably, the bumper portion is adapted to work along the impact axis, and the tongue portion extends along a tongue axis perpendicular to impact axis.
 - (11) Preferably, the first side wall extends along an axis Y_w perpendicular to the impact axis X_b and the tongue axis Z_t .
 - (12) Preferably, the service member further comprises a second side wall extending parallel to the first side wall on the opposite side of the tongue portion.
 - (13) Preferably, the service member further comprises a top wall extending over the tongue portion when looked along an axis Y_t going through the tongue portion parallel to the Y_w axis.
 - (14) Preferably, the top wall is inclined with respect to the tongue portion.
 - (15) Preferably, the service member further comprises a bottom wall extending below the tongue portion when looked along an axis Y_t going through the tongue portion parallel to the Y_w axis, the bottom wall being located on the opposite side of the tongue portion with respect to the top wall.
 - (16) Preferably, the bottom wall is inclined with respect to the tongue portion.
 - (17) Preferably, the service member further comprises a face wall perpendicular to the first side wall, wherein the tongue portion protrudes from the face wall.
 - (18) Preferably, the first side wall comprises a through-hole for a screw.
 - (19) Another object of the invention is an assembly comprising a service member as described and a vehicle cooling module, the vehicle cooling module comprising a slot and at least one contact side wall, the service member being attached to the vehicle cooling module so that the tongue portion is introduced into the slot and the first side wall is in contact with the contact side wall.
 - (20) Another object of the invention is an assembly comprising a service member as described and a vehicle cooling module, wherein the contact side wall comprises a through-hole for a screw coaxial with the through hole of the service member for fixing together the vehicle cooling module and the service member.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Examples of the invention will be apparent from and described in detail with reference to the accompanying drawings, in which:
- (2) FIG. 1a shows the first stage of the breaking of the mounting assembly and replacement by a service member according to the invention;
- (3) FIG. 1b shows the second stage of the breaking of the mounting assembly and replacement by a service member according to the invention;
- (4) FIG. 1c shows the third stage of the breaking of the mounting assembly and replacement by a service member according to the invention;
- (5) FIG. 2 shows the service member in a perspective view;
- (6) FIG. 3a shows the service member in top view, with FIG. 3b showing a closer view of the connection portion;
- (7) FIG. 4 shows another perspective view of the service member;
- (8) FIG. 5 shows cross-section view of the service member;
- (9) FIG. 6 shows a vehicle cooling module portion after removal of broken mounting assembly; and
- (10) FIG. 7 shows a vehicle cooling module portion with a service member.

DETAILED DESCRIPTION OF THE INVENTION

- (11) FIGS. 1a-1c show breaking of the mounting assembly of a vehicle cooling module 50 and

replacement by a service member **1** according to the invention. Once the vehicle hits an object, a force is transferred from a car beam by a crash bumper **61** to the vehicle cooling module **50**. A mounting element **62**, serving as attachment component of the vehicle cooling module **50** to vehicle's structure, breaks in a predetermined place. If the radiator is not damaged, the crash bumper **61** is removed, the broken area is polished off and a service member **1** is mounted on the cooling module. The details of the service member **1** will be explained below.

(12) FIG. **2** shows the service member **1** in a perspective view. In general, the service member **1** comprises a first element **10** with a bumper portion **11** and a second element **20** connected to the first element **10**. The function of the first element **10** is to receive impact from the car beam or other vehicle components. The first element **10** is also used for mounting the service member **1** to the vehicle cooling module **50**. The second element **20** is intended for mounting the service member **1** to the vehicle structure. What follows is that the second element **20** serves as a mounting component for the vehicle cooling module **50**.

(13) FIGS. **3a** and **3b** show the service member **1** in top view with a closer view on its connection portion **30**. The first element **10** is connected with the second element **20** by a connection portion **30** adapted to controllably break upon imparting a predetermined force to the bumper portion **11**. In particular, the connection portion **30** comprises a connecting arm **31** between the first element **10** and the second element **20**. Preferably, there is a plurality of connecting arms **31**. At least one of the arms **31** has a weakened portion **32** intended to break more easily than the rest of the connecting arms **31**. This can be realized by selectively reducing the thickness of the connecting arm **31**. In this case, there are three connecting arms **31** connecting the first and second elements **10**, **20**, wherein the middle one has a weakened portion **32**. Thanks to this weakened portion **32** it is possible to control the propagation of the crack line to run only through the connecting arms **31**, without going in the direction of either the first element **10** or the second element **20**.

(14) FIG. **4** shows another perspective view of the service member **1**, while FIG. **5** shows a cross-section view. On the first element **10** there is located a first fixing interface **12** for fixing the service member **1** to the vehicle module **50**, while on the second element **20** there is located a second fixing interface **22** for fixing the servicing member **1** to the vehicle structure.

(15) The first fixing interface **12** comprises a tongue portion **13**. The tongue portion **13** is adapted for releasable connection with a slot **51** in the vehicle cooling module **50**. In other words, the tongue portion **13** is shaped and dimensioned so that it can be inserted into a dedicated slot **51** in the vehicle cooling module **50** so that it is securely attached thereto, while retaining a possibility of detachment. Preferably, the tongue portion **13** is of plate-like shape and rectangular cross-section. The cross-section of the tongue portion **13** can be decreasing along its length for improved securing within the slot **51**. The tongue portion **13** can comprise a blocking section **14** adapted for frictional fixing in a slot **51** of a vehicle cooling module **50**. This blocking section **14** can be for example locally enlarged thickness.

(16) The first fixing interface **12** comprises a first side wall **15** extending adjacent to the tongue portion **13** between said tongue portion **13** and the bumper portion **11** so that a free space is created between the tongue portion **13** and the side wall **15**. In general, the bumper portion **11** is adapted to work along an impact axis X_b , and the tongue portion **13** extends along a tongue axis Z_t perpendicular to the impact axis X_b . The first side wall **15** extends along an axis Y_w perpendicular to the impact axis X_b and the tongue axis Z_t . Preferably, the first side wall **15** is a flat wall which extends from the tongue portion **13** substantially below the tongue portion **13**, preferably at a distance at least double the thickness of the tongue portion **13** measured along axis Y_t .

(17) As shown in FIG. **5**, there is a free space between the tongue portion **13** and the first side wall **15** when viewed along an axis X_t parallel to the impact axis X_b , the axis X_t going through the tongue portion **13**.

(18) The service member **1** can further comprise a second side wall **16** extending parallel to the first side wall **15** on the opposite side of the tongue portion **13**. Additionally, the service member **1**

can further comprise a top wall **17** extending over the tongue portion **13** when looked along an axis Yt going through the tongue portion **13** parallel to the Yw axis. The top wall **17** can be inclined with respect to the tongue portion **13**. There can be also a bottom wall **18** extending below the tongue portion **13** when looked along an axis Yt going through the tongue portion **13** parallel to the Yw axis, the bottom wall **18** being located on the opposite side of the tongue portion **13** with respect to the top wall **17**. This bottom wall **18** can be inclined with respect to the tongue portion **13**. The tongue portion **13** can protrude from a face wall **19** which is perpendicular to the first side wall **15**. In any case, there is a space created at least between the tongue portion **13** and the first side wall **15** into which a portion of the vehicle cooling module is introduced for secured connection which enables force transfer upon impact received through the bumper portion **11**. Because the force can be transferred securely, the connection portion **30** is allowed to break controllably. This particular design allows to achieve this in a simple and cost effective manner.

(19) The first side wall **15** can comprise a hole **19** for a screw, for example a through-hole, enabling more secure connection to the vehicle cooling module **50**.

(20) FIG. **6** shows the vehicle cooling module **50** portion after removal of broken mounting assembly. In the same perspective, FIG. **7** shows the vehicle cooling module **50** portion with the service member **1**. The vehicle cooling module **50**, in this case part of a radiator, has a slot **51** and at least one contact side wall **53**. When the service member **1** is attached to the vehicle cooling module **50**, the tongue portion **13** is introduced into the slot **51** and the first side wall **15** is in contact with the contact side wall **53**. In this manner, a secure arrangement is created between the cooling module **50** and the service member **1**. Preferably, the vehicle cooling module **50** comprises also a front contact wall **52** which after assembly is facing, and preferably is in contact, with the face wall **19** of the service member **1**. If there is a plurality of contact walls on the vehicle cooling module **50** as shown in FIG. **6**, the service member **1** can in fact encapsulate them and a correct positioning of the service member can be ensured after inserting the tongue portion **13** into the slot **51** of the vehicle cooling module **50**. The inclined top wall **17** and bottom wall **18** can follow the shape of the vehicle cooling module **50** for better alignment and protection of the connection. In FIG. **7**, the top wall **17** is not inclined.

(21) The contact side wall **53** can comprise a hole or a through-hole **54** for a screw coaxial with the through hole **19** of the service member **1** for fixing together the vehicle cooling module **50** and the service member **1** in a secured manner. In such case, a screw is inserted into both of the openings after assembly, and removed if needed for another replacement of the servicing member if another crash occurs.

(22) In any case, the service member **1** after assembly and upon crash will break within the connection area **30**. The components thereof can then be removed and another servicing member **1** can be mounted.

(23) Other variations to the disclosed embodiments can be understood and effected by those skilled in the art in practicing the claimed invention, from a study of drawings, the disclosure, and the appended claims. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to the advantage.

Claims

1. A servicing member for a vehicle cooling module, comprising a first element with a bumper portion, a first fixing interface located on the first element for fixing to the vehicle cooling module, a second element connected to the first element, a second fixing interface located on the second element for fixing the servicing member to a vehicle structure, wherein the first element is connected with the second element by a connection portion adapted to controllably break upon imparting a predetermined force to the bumper portion, wherein the first fixing interface includes a tongue portion, the tongue portion being adapted for releasable connection with slot in the vehicle

- cooling module, the first fixing interface including a first side wall extending adjacent to the tongue portion between said tongue portion and the bumper portion so that a free space is created between the tongue portion and the first side wall.
2. The servicing member according to claim 1, wherein the tongue portion is of plate shape and rectangular cross-section.
 3. The servicing member according to claim 1, wherein the tongue portion includes a blocking section adapted for frictional fixing in the slot of the vehicle cooling module.
 4. The servicing member according to claim 1, wherein the bumper portion is adapted to work along an impact axis, and the tongue portion extends along a tongue axis perpendicular to the impact axis.
 5. The servicing member according to claim 4, wherein the first side wall extends along an axis perpendicular to the impact axis and the tongue axis.
 6. The servicing member according to claim 1, further including a second side wall extending parallel to the first side wall on the opposite side of the tongue portion.
 7. The servicing member according to claim 4, further including a top wall extending over the tongue portion when viewed along an axis going through the tongue portion parallel to the axis perpendicular to the impact axis and the tongue axis.
 8. The service member according to claim 7, wherein the top wall is inclined with respect to the tongue portion.
 9. The servicing member according to claim 7, further including a bottom wall extending below the tongue portion when viewed along an axis going through the tongue portion parallel to the axis perpendicular to the impact axis and the tongue axis, the bottom wall being located on the opposite side of the tongue portion with respect to the top wall.
 10. The service member according to claim 9, wherein the bottom wall is inclined with respect to the tongue portion.
 11. The service member according to claim 1, further including a face wall perpendicular to the first side wall, wherein the tongue portion protrudes from the face wall.
 12. The service member according to claim 1, wherein the first side wall includes a through-hole for a screw.
 13. An assembly comprising: a servicing member for a vehicle cooling module, comprising a first element with a bumper portion, a first fixing interface located on the first element for fixing to the vehicle module, a second element connected to the first element, a second fixing interface located on the second element for fixing the servicing member to a vehicle structure, wherein the first element is connected with the second element by a connection portion adapted to controllably break upon imparting a predetermined force to the bumper portion, wherein the first fixing interface includes a tongue portion, the tongue portion being adapted for releasable connection with a slot in the vehicle cooling module, the first fixing interface including a first side wall extending adjacent to the tongue portion between said tongue portion and the bumper portion so that a free space is created between the tongue portion and the first side wall, and a vehicle cooling module, the vehicle cooling module including a slot and at least one contact side wall, the service member being attached to the vehicle cooling module so that the tongue portion is introduced into the slot and the first side wall is in contact with the contact side wall.
 14. The assembly according to claim 13, wherein the contact side wall includes a hole for a screw coaxial with the through-hole of the service member for fixing together the vehicle cooling module and the service member.
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